

Bangladesh Pesticide Policy Inventory

Global benchmark for safe & hazardous agrochemicals — dual-list regulatory model with full evidence base

EU ACTIVE SUBSTANCES	UNACCEPTABLE CO-FORMULANTS	VEGETABLE SAMPLES EXCEEDING MRL	HHP POISONING RATE
1,473	144	73%	900
977 banned · 421 authorized · 75 under review [28]	EU list, Reg. (EC) 2021/383 [+14 proposed] [14,15]	of the 29% found contaminated (n=1,577) [4]	per 100,000 people — among world's highest [1]

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KEY FINDINGS

Adopt the EU standard, not the US one.	EU approvals require a substance be proven safe; the US requires proof of unreasonable risk. This single difference explains why chlorothalonil is banned in the EU but still legal in Brazil. Bangladesh should follow the precautionary model. [28,30]
Ban is not enough — MRLs must fall too.	When the EU withdraws an active substance it cuts its MRLs to the limit of determination (0.01 mg/kg), which makes export use impossible. A de-registration without the MRL cut leaves a legal residue loophole. [43]
Codex compliance does not buy EU access.	For azinphos-methyl the US MRL is 1.5 ppm against the EU's 0.5 ppm; thiophanate-methyl is 2 ppm against 0.5 ppm. Meeting Codex alone leaves Bangladeshi produce rejected at the border. [37]
Co-formulants are over half the product.	Surfactants, solvents and other 'inerts' can exceed 50% of a formulation's weight and are frequently more toxic than the active ingredient. POE-tallowamines was long assumed safe, then banned. [15,33]
Residue failure is systemic, not incidental.	29% of 1,577 vegetable samples were contaminated and 73% of those exceeded MRLs. Gourd hit 100% contamination and the Hazard Quotient for cypermethrin in beans reached 263 for adults and 510 for adolescents. [4]
Regulation without alternatives will fail.	Eggplant shoot and fruit borer can destroy 100% of a crop and melon fruit fly up to 60% of cucurbit fruit. IPM capacity — traps, biopesticides, extension — must expand in step with the bans. [2,3]

Evidence references in [brackets] map to the Sources sheet. Figures are drawn from the source paper and reflect its stated scope; verify against primary regulators before policy adoption.

HHP Classification Criteria

The WHO/FAO eight-criterion test — a substance meeting ANY one condition is a Highly Hazardous Pesticide

THE EIGHT CRITERIA (FAO/WHO INTERNATIONAL DEFINITION)					
#	Criterion	Instrument	What it captures	Why it matters	Ref
1	Acute toxicity — WHO Class Ia or Ib	WHO Recommended Classification of Pesticides by Hazard	Ia = extremely hazardous; Ib = highly hazardous to humans on acute exposure	Worker poisoning risk	[17,18,26]
2	Carcinogenicity — GHS Category 1A or 1B	Globally Harmonized System (GHS)	Proven or presumed human carcinogen; captures chronic, non-lethal cancer risk	Long-latency disease burden	[18]
3	Mutagenicity — GHS Category 1A or 1B	Globally Harmonized System (GHS)	Proven or presumed mutagen; addresses genetic damage across generations	Population-level genetic risk	[18]
4	Reproductive toxicity — GHS Category 1A or 1B	Globally Harmonized System (GHS)	Proven or presumed reproductive toxin; covers fertility, foetal and lactation effects	Birth-defect and fertility risk	[18]
5	Listed in Stockholm Convention Annex A or B	Stockholm Convention on POPs	Persistent, bioaccumulative, widely distributed POP — Annex A eliminates, Annex B restricts	Environmental persistence	[18,28]
6	Listed in Rotterdam Convention Annex III	Rotterdam Convention on Prior Informed Consent (PIC)	Banned or severely restricted by at least two Parties for health or environmental reasons	International consensus to restrict	[18,28]
7	Listed under the Montreal Protocol	Montreal Protocol on Ozone-Depleting Substances	Affects the ozone layer; a marker of unusually high environmental hazard	Ozone-layer depletion	[18]
8	Severe or irreversible local harm at use conditions	FAO/WHO plus International Code of Conduct on Pesticide Management	Demonstrable high incidence of severe/irreversible effects under actual local use	National discretion to act	[18,24]

Note: the source paper's narrative numbers Montreal as criterion 6 and omits the seventh. Reconstructing from its own statement that severe local harm is the 'eighth' criterion places carcinogenicity, mutagenicity and reproductive toxicity at 2, 3 and 4 respectively, which restores the complete eight.

TREATY INVENTORIES BEHIND CRITERIA 5-7				
Instrument / List	Legal effect	Context	Scale	Ref
Stockholm Convention — Annex A	Production and use must be eliminated worldwide	39 POPs, of which 23 are pesticides. Chlorpyrifos is listed here and is consequently auto-banned in the EU.	39 POPs / 23 pesticides	[28,30]
Rotterdam Convention — Annex III	PIC procedure; trade requires prior informed consent	Inclusion requires a ban or severe restriction by at least two Parties for health or environmental reasons.	57 chemicals / 38 pesticides	[28]
EU unacceptable co-formulants	Banned in all plant protection products	Established under Reg. (EC) 2021/383 amending Annex III to Reg. (EC) 1107/2009.	144 substances	[14,15,33]
EU proposed additions (Dec 2025)	High carcinogenicity, mutagenicity, reproductive toxicity	Commission adoption targeted Q1 2026, followed by a two-year withdrawal period for Member States.	+14 substances	[14,44]

Active Ingredients Inventory

Dual-list model: EU-derived whitelist, WHO/FAO-derived blacklist

APPROVED & LOW-RISK (WHITELIST)						
Active Ingredient	Group/Class	Regulatory Status	Primary Use	Bangladesh Relevance		Ref
Acetic Acid (Vinegar)	Basic Substance	EU Basic / US 25(b)	Fungicide/Bactericide	Broad-spectrum, safe for vegetables		[8,10]
Sucrose	Basic Substance	EU Low-Risk	Attractant/Adjuvant	Safe formulation base		[7,19]
Ascorbic Acid (Vit C)	Basic Substance	EU Low-Risk	Antioxidant	Safe post-harvest		[7,8]
Beeswax	Basic Substance	EU Low-Risk	Coating/Protectant	Fruit coating; CAS 8012-89-3		[7,19]
Calcium	Basic Substance	EU Low-Risk	Nutrient/Fungicide	Blossom end rot in tomatoes		[7,8]
Bacillus thuringiensis (Bt)	Biopesticide	EU/US Approved	Insecticide	Eggplant shoot borer control		[29]
Neem Oil (Azadirachtin)	Botanical	EU/US Approved	Insecticide	Broad pest control (IPM)		[29]
Spinosad	Botanical	EU/US Approved	Insecticide	Bean pod borer; listed alternative		[29]

EU low-risk approval carries a 15-year inclusion period versus the standard 10, and basic substances receive indefinite EU-wide authorisation — both strong incentives to register safer active ingredients. [8,10]

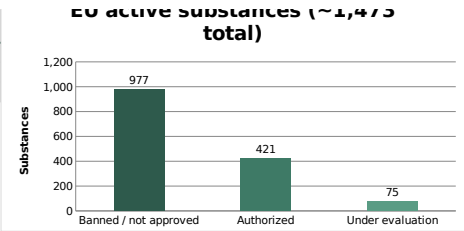
HIGHLY HAZARDOUS PESTICIDES (BLACKLIST)							
Active Ingredient	Group/Class	Hazard Classification	Global Status	HHP Criteria Met	Policy Evidence	Bangladesh Policy Action	Ref
Chlorpyrifos	Organophosphate	Neurotoxic; POP	Banned EU; Annex A Stockholm	5, 8	Fails EU approval; MRL cut to 0.01 mg/kg	Immediate Ban & Phase-out	[30,39]
Paraquat	Bipyridyl	Acute toxicity (WHO Ia)	Banned in 72 countries incl. Brazil, China	1, 8	Among the most prevalent actives in Bangladeshi use	Immediate Ban	[27]
Monocrotophos	Organophosphate	WHO HHP; Rotterdam listing	Banned globally	1, 6	Still in widespread domestic use	Immediate Ban	[1]
Dichlorvos	Organophosphate	Carcinogen; WHO HHP	Restricted / being phased out	2, 3, 4	Top-5 active ingredient by Bangladeshi use	Immediate Ban	[1]
Carbofuran	Carbamate	High acute toxicity; WHO HHP	Banned EU/US	1, 8	Drives stem borer control in paddy	Immediate Ban	[1]
Mancozeb	Dithiocarbamate	Endocrine disruptor; reproductive toxin	EU renewal denied	4, 8	Principal blight fungicide on tomato	Phase-out & Substitute	[30]
Diazinon	Organophosphate	Neurotoxic; environmental toxicity	Restricted	1, 5, 8	Frequently detected above MRL in vegetables	Phase-out & Substitute	[4,46]
Carbaryl	Carbamate	Carcinogen (GHS 1B)	Restricted; US tolerances established	2, 8	Legacy organophosphate with no EU approval	Phase-out	[39]
Chlorothalonil	Organophthalate	Non-EU-approved fungicide	Banned EU; still legal in Brazil	8	Illustrates EU/US regulatory divergence directly	Immediate Ban	[28]
Buprofezin	Growth Regulator	Suspected endocrine disruptor	Draft EU non-approval, final 2026	8	Non-renewal expected to trigger an MRL cut	Pre-emptive Ban	[44]
Flufenacet	Chloroacetamide Herbicide	Endocrine disruption (2024 EFSA)	Not re-approved EU	8	Withdrawn specifically on endocrine-disruption grounds	Immediate Ban	[43]
Prochloraz	Imidazole Fungicide	Reproductive toxin 1B; suspected ED	EU candidate for substitution	4, 8	Named substitution candidate alongside epoxiconazole	Phase-out	[30]
Vinclozolin	Dicarboximide Fungicide	Endocrine disrupting	Banned in the EU for years	2, 3, 4, 8	Long-standing endocrine case precedent	Immediate Ban	[30]
Malathion	Organophosphate	Neurotoxic; frequent residue detection	Restricted	1, 5, 8	Top-5 active ingredient by Bangladeshi use	Phase-out & Substitute	[1,4]
Cypermethrin	Pyrethroid	High Hazard Quotient in residue testing	EU restricted, not approved	8	HQ 263 adults / 510 adolescents measured in beans	Phase-out synthetic pyrethroids	[4,28]
Chlordane	Organochlorine	Persistent legacy insecticide; no US tolerance	US sets an action level, not a tolerance	2, 3, 4, 8	An action level instead of a tolerance signals long persistence and legacy use — a strict-control candidate	Strict control — candidate for prohibition	[30,38]

WHERE THE THREE SYSTEMS DISAGREE — CROSS-JURISDICTION STATUS					
Active Ingredient	WHO Acute Toxicity Class	EU Status	US EPA Tolerance Status	Stockholm Convention Status	Rotterdam Convention Status
Chlorpyrifos	Not specified in source	Banned / Not Approved [30]	Tolerances established [39]	Listed as a POP (Annex A) [30]	Not listed in provided sources
Carbaryl	Not specified in source	Not specified in source	Tolerances established [23]	Not listed in provided sources	Not listed in provided sources
Mancozeb	Not specified in source	Not Approved (renewal denied) [30]	Tolerances established [42]	Not listed in provided sources	Not listed in provided sources
Paraquat	Not specified in source	Not specified in source	Tolerances established [27]	Not listed in provided sources	Not listed in provided sources
Chlordane	Not specified in source	Not specified in source	No tolerance; action level set [38]	Not listed in provided sources	Not listed in provided sources

Chlorpyrifos is the clearest divergence: banned in the EU, an Annex A POP, yet still carrying established US tolerances. Chlordane is the inverse — no US tolerance at all, only an action level, which signals long persistence and legacy use. "Not specified in source" means the source paper did not state a value, not that no status exists. [30,38,39,42]

EU Status	Substances		Share	Role in national policy	Ref
Banned / not approved	977	66.3%		Drives the national blacklist	[28]
Authorized	421	28.6%		Defines the eligible whitelist	[28]
Under evaluation	75	5.1%		Watchlist — monitor for withdrawal	[28]

'HHP Criteria Met' maps each substance to the numbered WHO/FAO tests on the HHP_Criteria sheet. Two-thirds of EU-evaluated actives are non-approved — adopting the EU list as the national whitelist is the single highest-leverage policy decision available. [28]



Formulation Materials (Inerts & Co-formulants)

'Inert' is a misnomer — co-formulants exceed 50% of product weight and are regulated in their own right

SAFE & EXEMPT CO-FORMULANTS					
Substance Name	CAS Number	Function	Regulatory Status	Notes	Ref
Acetic Acid, Sodium Salt	127-09-3	Solvent / Adjuvant	US tolerance-exempt	Safe for food-use sites	[5]
Alpha-cyclodextrin	10016-20-3	Solubilizer / Carrier	US tolerance-exempt	Carbohydrate-based carrier	[5]
Citric acid, disodium salt	144-33-2	Chelating agent / pH	US tolerance-exempt	Common food additive	[5]
Carbonic acid, monopotassium salt	298-14-6	pH Adjuster	US tolerance-exempt	Potassium bicarbonate equivalent	[5]
Ethanol	64-17-5	Solvent	GRAS for food use	Universal safe solvent	[5]
Lactic acid, ethyl ester	97-64-3	Solvent	US tolerance-exempt	Biodegradable solvent	[5]
Silica, amorphous, fumed	112945-52-5	Anti-caking agent	US tolerance-exempt	Inert dust / powder base	[5]
Beeswax	8012-89-3	Emulsifier / Thickener	EU low-risk; US exempt	Dual-listed safe co-formulant	[5,7,19]
Sucrose	57-50-1	Adjuvant / Nutrient	EU low-risk	EU-approved basic substance	[7]

UNACCEPTABLE & HAZARDOUS CO-FORMULANTS					
Substance Name	CAS Number	Function	Hazard Profile	Regulatory Status & Policy Note	Ref
POE-tallowamines	61791-26-2	Surfactant	Aquatic toxicity; endocrine active	EU List 1 — banned. Long assumed safe, later found as toxic as an active ingredient: the defining oversight failure.	[15,33]
Acetaldehyde	75-07-0	Solvent / Intermediate	Carcinogenic; mutagenic	Proposed ban (Dec 2025) — one of 14 co-formulants proposed for prohibition.	[14,44]
Propylene oxide	75-56-8	Intermediate / Solvent	Carcinogenic (GHS 1B)	Proposed ban (Dec 2025) — high carcinogenicity drove the proposal.	[14]
D4 / D5 (Cyclotetrasiloxane / Cyclosiloxane)	107-52-8 / 541-02-6	Solvent / Carrier	PBT; vPvB; persistent	Proposed ban (Dec 2025) — persistent, bioaccumulative siloxanes.	[14]
D6 — Dodecamethylcyclohexasiloxane	107-54-7	Solvent / Carrier	PBT / vPvB	Proposed ban (Dec 2025) — highest-molecular-weight member of the siloxane series.	[14]
1,4-dioxane	123-91-1	Solvent impurity	Carcinogenic	Proposed ban (Dec 2025) — named explicitly in the 2025 proposal.	[14,44]
Benzotriazole derivatives	Not available	Antioxidant / Stabilizer	High toxicity; PBT	Proposed ban (Dec 2025) — stabiliser class with no acceptable safety case.	[14]

MANDATORY DISCLOSURE REQUIREMENTS			
Requirement	Rule	Rationale	Ref
Full ingredient naming	Every formulation ingredient must be disclosed on the product label by name and concentration — not merely as a total percentage of the formula.	Regulators, clinicians and applicators cannot otherwise assess real exposure.	[40]
Terminology change	'Inert' must be replaced with 'other' on all labels; the term implied biological harmlessness that the evidence does not support.	The EPA made this change for exactly this reason.	[40]
Burden-of-proof shift	Where a company fails to supply toxicity data for a co-formulant, registration must be refused and existing authorisations suspended.	EU General Court ruling shifts the burden of proof away from regulators.	[33,35]
Unknown toxicity	Substances with insufficient data (US List 3 equivalent) require a full safety dossier before any product containing them is registered.	Applies the precautionary principle to the ~1,700 chemicals with no toxicity data.	[25,40]

Co-formulant Tiering Model

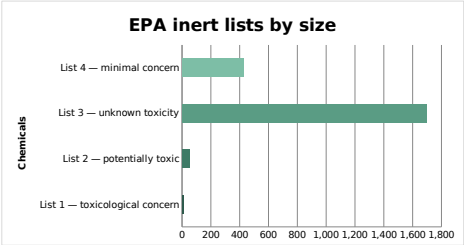
US EPA toxicity Lists translated into a proposed Bangladeshi red / amber / green scheme

US EPA INERT INGREDIENT LISTS (SOURCE MODEL)						
EPA List	Name	Count	Hazard basis	Handling under US rules	Proposed Bangladesh status	Ref
List 1	Inerts of Toxicological Concern	8	Carcinogenic, neurotoxic, or causing adverse developmental/reproductive or other chronic effects	Must be named individually on the product label	RED — Prohibited	[25,40]
List 2	Potentially Toxic Inerts	52	Structurally similar to List 1, or data suggest toxicity; high testing priority	Precautionary controls pending full assessment	AMBER — Restricted	[25,40]
List 3	Inerts of Unknown Toxicity	1,700	Insufficient data for the EPA to determine toxicity; assessments under development	Already prohibited in organic production	AMBER — Data required before registration	[20,25,40]
List 4A	Inerts of Minimal Concern	430	Considered of minimal concern (List 4 total, subdivided into 4A and 4B)	No restriction	GREEN — Permitted	[19,40]
List 4B	Inerts with Sufficient Safety Data	n/a	EPA concluded current use patterns will not adversely affect public health or the environment	No restriction	GREEN — Permitted	[40]

PROPOSED NATIONAL COLOUR SCHEME					
Tier	Scope	Operative rule		Rationale	Ref
RED — Prohibited	Mirrors the EU's 144-substance unacceptable list plus the 14 proposed additions; also US List 1.	Absolute ban on import, sale and use in any agricultural product.		Known carcinogenic, mutagenic, reproductive toxin or POP status.	[14,15,40]
AMBER — Conditional	Mirrors US List 2 and List 3 — substances of concern or of unknown toxicity.	Permitted only with a full safety dossier; special handling and use limitations apply.		Precautionary principle applied where data are absent, rather than presumed benign.	[25,40]
GREEN — Permitted	Mirrors US List 4A/4B — minimal concern with supporting data.	Unrestricted, subject to label disclosure of name and concentration.		430+ inerts have already been cleared on this basis.	[19,40]

TIER DISTRIBUTION			
EPA List tier	Chemicals		Handling
List 1 — toxicological concern	8	Named on every label	
List 2 — potentially toxic	52	High testing priority	
List 3 — unknown toxicity	1,700	Data required before registration	
List 4 — minimal concern	430	No restriction	

List 3 is roughly four times larger than List 1 and List 2 combined. A national tiering scheme is only meaningful if unknown-toxicity inerts are gated behind a data dossier — otherwise 1,700 unassessed chemicals enter the supply chain by default. [25,40]



Comparative Regulatory Frameworks

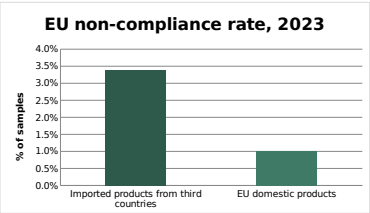
EU vs US EPA vs Codex Alimentarius — no unified global standard exists

Side-by-Side Comparison				
Feature	European Union	United States (US EPA)	Codex Alimentarius	Ref
Governing instrument	Reg. (EC) No 1107/2009 (approval); Reg. (EC) No 396/2005 (MRLs)	FIFRA (registration); FFDCa (tolerances)	Not a regulation — FAO/WHO food standards	[30,37,42]
Core philosophy	Precautionary: must prove NOT toxic	Risk management: must establish no unreasonable risk	Trade facilitation and consumer health protection	[28]
Approval approach	Centralised and harmonised across all member states since 2008	Decentralised, state-by-state registration	Not applicable — sets standards for national regulators	[28,37]
Handling of inerts	Comprehensive regulation; 144-substance unacceptable co-formulant list	Tiered classification — Lists 1, 2, 3, 4	Guidelines for compounds of low public health concern	[15,22,25,40]
Chlorpyrifos status	Banned / not approved (Annex A POP)	Tolerances established for food uses	MRLs exist but subject to review	[30,39,42]
Public data access	High — official databases and reports	Moderate — databases and notices online	High — MRL database publicly available	[9,10,11,34,40]
Net effect for Bangladesh	Adopt as primary benchmark: most stringent, precautionary, export-aligned	Useful as a ceiling and a data source, not as a safety floor	Baseline for trade; insufficient for EU market access	[28,29,37]

MRL Divergence — Why Codex Compliance is Not Enough						
Pesticide (apple MRL study)	US (ppm)	EU (ppm)	Codex (ppm)	US:EU	Implication	Ref
Azinphos-methyl	1.5	0.5	0.5	3.0x	US tolerance permits 3x the residue the EU allows	[37]
Thiophanate-methyl	2.0	0.5	0.5	4.0x	4x differential; drives border rejection risk	[37]

EU Non-Compliance Rate, 2023			
Product category	Non-compliance rate	Note	Ref
Imported products from third countries	3.4%	Main sources of non-compliant goods: Turkey, India, Egypt	[42]
EU domestic products	1.0%	Internal baseline for comparison	[42]

Imported produce fails EU residue testing 3.4x more often than EU domestic produce. Bangladesh must test against the strictest applicable MRL, not the Codex baseline, or face rejection at the border. [42]



Bangladesh Baseline Evidence

National contamination, poisoning and residue data underpinning the policy case

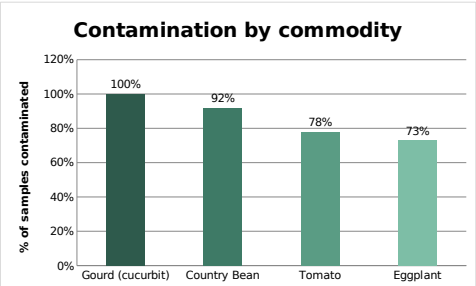
NATIONAL INDICATORS				
Indicator	Value	Unit	Context	Ref
Pesticide use, 2000	19,000	metric tonnes	Baseline year of the reported doubling	[1]
Pesticide use, 2022	37,000	metric tonnes	More than doubled in 22 years	[1]
Insecticide poisoning rate	900	per 100,000 people	Among the highest reported worldwide	[1]
Farmers among cancer patients	33	% of hospital cancer diagnoses	Points to chronic exposure effects	[1]
Farmers using little or no PPE	87	% of surveyed vegetable farmers	Near-universal disregard for protection	[1]
Vegetable production	20	million metric tonnes	World's third-largest producer	[1]
Rice production	39	million tonnes annually	World's fourth-highest producer	[3]

VEGETABLE RESIDUE MONITORING — DECADE REVIEW				
Measure	Value	Unit	Interpretation	Ref
Total samples analysed	1,577	samples	Ten-year comprehensive review of Bangladeshi vegetables	[4]
Contaminated samples	29	% of all samples	461 of 1,577 samples carried detectable residues	[4]
Exceeded MRL	73	% of contaminated samples	Systemic failure to respect pre-harvest intervals	[4]
Hazard Quotient — cypermethrin in beans, adults	263	HQ (threshold 1)	Chronic health risk far above threshold	[4]
Hazard Quotient — cypermethrin in beans, adolescents	510	HQ (threshold 1)	Most exposed group affected most severely	[4]

CONTAMINATION RATE BY COMMODITY				
Commodity	Contaminated samples	Note		Ref
Gourd (cucurbit)	100%	Every sample tested was contaminated		[4]
Country Bean	92%	Highest detection after gourd		[4]
Tomato	78%	Mancozeb-driven blight control is implicated		[4]
Eggplant	73%	Shoot and fruit borer spray regime implicated		[4]

MOST FREQUENTLY DETECTED PESTICIDES				
Chlorpyrifos	Dimethoate	Diazinon	Malathion	

All four are organophosphorus compounds with established neurotoxic effects. Three of the four — chlorpyrifos, diazinon and malathion — appear on this workbook's blacklist. [4,46]



Bangladesh Context & Policy Matrix

Crop-by-crop transition plan with yield-loss and residue context

CROP-SPECIFIC TRANSITION PLAN						
Target Crop	Major Pest Pressure	Current Hazardous Use	Recommended Safe Alternative	Policy Directive	Loss / Residue Context	Ref
Rice (Paddy)	Stem borer, brown planthopper, rice blast, tungro virus	Carbofuran, Diazinon	Bt, Neem, IPM traps; disease-resistant seed	Ban HHPs, subsidise IPM	Up to 62% yield loss from five major diseases; 39 Mt/yr production	[1,3]
Eggplant	Shoot & Fruit Borer	Dichlorvos, Monocrotophos	Bt sprays, pheromone traps	Strict MRL enforcement	Borer can cause 100% crop loss; 73% of samples contaminated	[2,4]
Cucurbits (Gourd)	Melon Fruit Fly	Cypermethrin, Malathion	Pheromone traps, biopesticides	Phase-out synthetic pyrethroids	Melon fruit fly destroys up to 60% of fruit; 100% contamination	[2,4]
Country Bean	Bean Pod Borer	Chlorpyrifos	Neem, Bt, Spinosad	Immediate chlorpyrifos ban	92% contaminated; cypermethrin HQ 263 adults / 510 adolescents	[2,4]
Tomato	Blight, pest complex	Mancozeb	Copper-based (low risk), Bt	Mancozeb phase-out	78% of samples contaminated; EU renewal denied	[3,4,30]
Okra, Cabbage, Cauliflower	Variable pest complex	Residual OP / pyrethroid use	Neem, Bt, Spinosad, IPM	Extend dual-list model	Part of national vegetable output of 19.7 Mt	[1,2]
Wheat, Jute	Weeds, stem borers	Residual broad-spectrum use	Bt, mechanical control	Extend dual-list model	Two of the three dominant focus crops alongside rice	[3]

FIVE-PILLAR POLICY FRAMEWORK				
#	Pillar	Action	Expected effect	Ref
1	Blacklist HHPs	Automatically prohibit any substance meeting any of the eight WHO/FAO criteria, plus the 144 unacceptable co-formulants and 14 proposed additions.	Removes the most dangerous chemicals from the agricultural supply chain	[14,15,18]
2	Whitelist actives	Permit registration only for substances on the EU approved list, reviewed regularly in consultation with scientific experts.	Highest achievable safety bar; aligns with export markets	[9,28]
3	Restructure inerts	Adopt EPA-style tiering; ban red-list co-formulants, gate unknown-toxicity substances behind a safety dossier, mandate full label disclosure.	Closes the co-formulant loophole that voids otherwise-safe actives	[25,40]
4	Residue monitoring	Build a national monitoring programme modelled on the EU's annual effort; test against Codex and, where stricter, EU MRLs; publish all results.	73% of contaminated samples currently exceed MRLs	[4,29,42]
5	Fund alternatives	Increase IPM research, extension and subsidies — pheromone traps, biopesticides, beneficial insects; update the national agro-input list.	Without viable alternatives, bans reduce yield and are ignored	[2,3,29]

Pillars 1–3 are regulatory and can be enacted administratively. Pillars 4–5 determine whether they hold — enforcement without IPM capacity shifts growers toward unregistered and more hazardous products.

Sources

46 references underpinning every figure in this workbook

REFERENCE LIST		
Ref	Source	URL
[1]	Pesticide safety behavior among vegetable farmers in Bangladesh	https://pmc.ncbi.nlm.nih.gov/articles/PMC11721242/
[2]	Vegetable IPM in Bangladesh	https://ipmworld.umn.edu/rahman
[3]	Prevention is best defense against Bangladesh crop diseases	https://news.mongabay.com/2024/01/prevention-is-best-defense-against-bangladesh-crop-diseases-researchers-say/
[4]	Pesticides in vegetable production in Bangladesh	https://www.sciencedirect.com/science/article/pii/S2214750023000951
[5]	40 CFR 180.950 — Tolerance exemptions for minimal risk pesticides	https://www.law.cornell.edu/cfr/text/40/180.950
[6]	EPA Revises Minimum Risk Pesticide Exemption Regulations	https://www.lawbc.com/epa-revises-minimum-risk-pesticide-exemption-regulations/
[7]	Low Risk Plant Protection Products (European Commission)	https://food.ec.europa.eu/system/files/2017-09/pesticides_sup_low-risk-ppps.pdf
[8]	Low-risk plant protection products (CTGB)	https://english.ctgb.nl/topics/working-together-on-sustainable-use/low-risk-products/low-risk-plant-protection-products
[9]	EU Pesticides Database (European Commission)	https://food.ec.europa.eu/plants/pesticides/eu-pesticides-database_en
[10]	Basic substances (BAES)	https://www.baes.gv.at/en/admission/plant-protection-products/basic-substances
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Source document: "Safeguarding Bangladesh: A Global Benchmark for Banning Hazardous Pesticides and Building a Safer Inventory". Reference numbers are preserved from that document so every citation remains traceable to it.